

# Introduction to Artificial Intelligence and RAG

Artificial Intelligence (AI) is a field of computer science that focuses on building systems capable of performing tasks that normally require human intelligence.

Machine Learning (ML) is a subset of AI in which computers learn patterns from data and use those patterns to make predictions or decisions.

Generative AI refers to models that can create new content such as text, images, audio, video, and code. Large Language Models (LLMs) are commonly used for generating and understanding text.

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. Relevant information is retrieved from documents and supplied to the model as context before the model generates an answer.

A typical RAG pipeline contains document loading, text splitting, embedding generation, vector storage, similarity search, context retrieval, and answer generation.